Scenario 3 Sociocultural Data Biography (SDB): Historical Maryland Domestic Traffic Advertisements Corpus

This Sociocultural Data Biography (SDB) details the social, institutional, legal, and transactional mechanisms that generated the archival records preserved in the primary corpus sample. All frequency analyses and quantitative patterns cited here are derived from the sample dataset Sample of Crawled Domestic Traffic Ads- Corrected.csv.

1. Contextual Adequacy — Systemic Contextual Analysis

(1) Record-Generating Purpose and Institutional Mandate
The records in the primary corpus were generated as paid public advertisements in early American commercial newspapers. Their primary function was commercial matchmaking: facilitating the sale or hiring of enslaved labor. Because publication incurred costs per word or line, the text was optimized exclusively to satisfy the business interests of the advertiser. Information was recorded only if it enhanced the market value of the enslaved person (e.g., highly specific vocational details in Skills Specified), or if it protected the transaction from legal liability or post-sale disputes. Details irrelevant to market value or contract enforcement were systematically excluded, creating a structural omission of the non-economic, personal, or familial dimensions of the enslaved individuals' lived experiences.

(2) Socio-Legal Structures Governing Maryland Slave Transactions
Socio-legal codes in Maryland heavily regulated the internal slave trade to mitigate economic liabilities and control the free and enslaved Black populations. As documented in Christopher Haley's *"Legacy of Slavery in Maryland: 'Chattels'"* (Journal of Slavery and Data Preservation, 2024), chattel records at the Maryland State Archives—including bills of sale, chattel mortgages, and releases—reveal that enslaved people were legally classified alongside crops, livestock, and wagons.

Furthermore, specific legislative acts restricted manumission and out-of-state trafficking. Under the Maryland Act of 1752 (documented in *Record #17: An Act to prevent disabled and superannuated Slaves being set free*), enslavers were legally barred from manumitting elderly or infirm enslaved individuals by last will or testament to prevent them from becoming financial burdens on the local parish. This law was repeatedly renewed throughout the 18th century (e.g., *Record #16, 1755* and *Record #15, 1758*).

By the 19th century, the state intensified its legal borders. In the landmark *Act of 1831, Chapter 323 ("An act relating to Free Negroes and Slaves")*, Maryland implemented strict restrictions on

the movement and legal status of Black individuals. Subsequent supplements, such as the Act of February 12, 1833 (*Record #25*), clarified exemptions for residents who moved enslaved labor across state borders out of agricultural necessity, while the Act of March 18, 1840 (*Record #27*), added further compliance burdens. These legal frameworks required advertisers to document specific details in the text, such as a clear statement of title or term of service, to shield buyers and sellers from regulatory infractions or fraud.

(3) Actors, Perspectives, and Authority

The authorial locus of the record is captured in the subscriber fields: Subscriber Firstname, Subscriber Lastname, Subscriber County, Subscriber Locale, and Subscriber State. The data reflects exclusively the perspective, economic motivations, and racial classifications of these white enslavers and commercial agents. Completely excluded from this index are the voices, perspectives, and self-identities of the enslaved individuals themselves. Enslaved persons appear strictly as passive objects of transaction rather than authors of their historical narratives.

(4) Representational Conventions

The corpus employs highly racialized and commodifying descriptive language within the FULL TEXT, Skills Specified, and Age Specified fields. Enslaved human beings are systematically translated into abstract commercial items using descriptive terms like "valuable," "stout," "healthy," or "sober." These descriptions must be interpreted as rhetorical and strategic constructions designed to maximize sales or leasing rates rather than objective biographical profiles. Physical descriptions and age estimates served as standard commercial markers of capital valuation and labor capacity within an antebellum economy that capitalized human bodies.

(5) Spatial and Temporal Variation

Transactional practices and corpus composition exhibit significant variation across space and time. Spatially, the fields County, Sale Location County, Sale Location Locale, and Sale Location State document distinct regional labor regimes across Maryland. For example, records originating from the Eastern Shore (e.g., Talbot County or Dorchester County) reflect an agricultural landscape transitioning away from tobacco cultivation, as explored in Jennifer Hull Dorsey's *Hirelings: African American Workers and Free Labor in Early Maryland* (2011), which traces how manumitted and freeborn African Americans refashioned the region's economy as wage laborers alongside declining slave systems.

Temporally, the corpus tracks fluctuations across the antebellum period through three distinct fields: Ad Date, Press Date, and Sale Date. In this sample, Ad Date represents the day the advertiser drafted or submitted the text, while Press Date represents the actual printing date of the newspaper issue. Analysis of the sample indicates that Ad Date and Press Date often match or Ad Date slightly precedes Press Date by a variable interval of days depending on weekly printing schedules. However, because documentation for the precise technical collection pipeline is incomplete, the exact operational divergence between these two fields remains an open interpretive question. Sale Date is less frequently populated, reflecting a structured absence that

depends on whether an auction was scheduled for a specific calendar date or negotiated privately.

(6) Structured Absences, Silences, and Uncertainties

Omissions within the corpus are rarely accidental; they are primarily structured by the commercial nature of the source material:

- **Anonymization:** The fields First Name and Last Name are frequently blank or contain placeholders like "boy" or "negro man." This deliberate omission reflects the strategic choice of the advertiser, who prioritized broad demographic or vocational markers ("excellent carriage driver," "good washer and ironer") over individual identities.
- **Transaction Incompleteness:** The fields Sale Date and Sale Disposition are heavily unpopulated. Because newspapers only recorded the *initiation* of a commercial intent (the advertisement) and not its final resolution, the ultimate fate of the enslaved individual remains an unrecorded administrative silence.
- **Unstructured Detail:** Dense biographical notes and specific transactional conditions are concentrated in the unstructured text of FULL TEXT and Notes rather than parsed into separate columns, posing a clear challenge for automated categorical extraction.

Intersection of Dimensions

A prominent feature of this dataset is the high concentration of specific vocational attributes recorded in Skills Specified for young males (aged 15–25) alongside a near-total absence of family surnames in Last Name. This specific configuration cannot be explained by any single dimension. Instead, it is produced by the intersection of **Dimension 1 (Commercial Function)** and **Dimension 4 (Representational Conventions)**: the advertiser maximized market utility by explicitly highlighting valuable skills ("groom," "waiter," "carriage driver"), while simultaneously stripping away family markers to reinforce chattel status and facilitate the individual separation of family members.

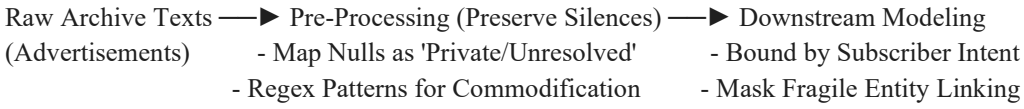
2. Evidentiary Adequacy — Source Material Domain Evaluation

Every systemic claim made in the context section is directly traceable to the primary corpus or specific historical documentation:

- **Legal Controls and Chattel Status:** The claim that enslaved human beings were legally grouped with livestock and tools is grounded in Christopher Haley's dataset article, *"Legacy of Slavery in Maryland: 'Chattels'"* (Journal of Slavery and Data Preservation, 2024, pp. 36-40), which notes that the dataset is derived from county chattel sales records that enumerated crops, livestock, and farm implements alongside enslaved persons.
- **Superannuation Restrictions:** The restrictions on freeing elderly slaves are supported by the Maryland Acts of Assembly from 1752, 1755, and 1758 (*Enduring Connections*, Nabb Research Center, Records #15, #16, and #17), which explicitly show a legislative mandate "to prevent disabled and superannuated Slaves being set Free."

- **Advertiser Subscriptions:** The assertion that the subscriber is the sole authorial voice is verified by the structure of the primary CSV sample, where rows terminate with references to specific local agents, such as the row entry containing "Goldsborough & Henry, Agents" in the Cambridge Democrat newspaper.
- **Uncertainty in Age and Skills:** The fields Age Specified and Skills Specified are highly contingent. For example, entries listing an individual as "about 25 years of age" or an "excellent cook" reflect unverified, strategic assertions by the seller. These figures must be treated as market claims rather than verified biological or historical data points.

3. Translational Adequacy — Data Pre-Processing and Analytical Design Guidance



(1) Data Preparation and Representation

- **Handling Commodifying Language:** Computational text processing should not smooth out or strip hyper-commercialized adjectives (e.g., "valuable," "stout") from FULL TEXT. Instead, these terms should be extracted via targeted regular expressions to construct an explicit computational proxy for "rhetorical valuation."
- **Encoding Missing Values:** Missing entries in Sale Date and Sale Disposition must not be imputed or treated as missing data errors. They must be encoded as a distinct, substantive categorical state: Unresolved_Private_Transaction. This preserves the historical fact that newspapers recorded commercial intentions rather than completions.

(2) Concept Operationalization

- **Labor Commodification:** To operationalize the concept of "labor commodification" as a computational construct, researchers must analyze the linguistic proximity between technical vocations in Skills Specified and financial metrics within FULL TEXT. Because this concept is embedded primarily in unstructured fields, researchers should deploy token-distance metrics to calculate how closely an individual's physical descriptions are bound to terms indicating leasing periods or dollar amounts.

(3) Analytical Design

- **Inappropriate Analysis (Entity Resolution):** Attempting automated entity resolution or biographical reconstruction across records using only First Name and County is methodologically unsafe. Due to the systemic and strategic anonymization practiced by advertisers, matching an individual named "Tom" across multiple ads will generate false collations, treating distinct individuals as a single historical entity.

(4) Method and Algorithm Selection

- **Algorithmic Bias and Bounds:** Any text-mining or topic-modeling algorithm applied to this dataset will inevitably map the cognitive classifications and market logic of the white enslavers (Subscriber fields). The output configurations do not represent the actual distribution of Black labor or life in Maryland, but rather the distribution of what white advertisers found profitable to publicize.

(5) Explicit Acknowledgment of Ambiguity

A clear translational conflict exists regarding how to encode the spatial attribute County when it diverges from Sale Location County. For instance, an ad may be placed by a subscriber residing in Dorchester County, but the physical auction is scheduled to take place at a courthouse in Talbot County. The context supports two equally defensible computational mappings:

- *Option A:* Encode the record based on the subscriber's location to map the geography of slave-owning capital.
- *Option B:* Encode the record based on the sale location to map the physical geography of the domestic traffic.

Because both options are historically justifiable, researchers must document this ambiguity and maintain dual spatial indexes rather than collapsing them into a single geographical coordinate.

4. Analytical Lifecycle Utility — Interpretive Guidance for the Research Lifecycle

This section provides interpretive frameworks for evaluating computational anomalies discovered during downstream research.

Retrospective Evaluation of Computational Outputs

Hypothetical Scenario A: The Discovery of a Null Correlation

- *Plausible Computational Output:* A researcher fits a linear regression model to test the impact of age (Age Specified) on the text length of the advertisement (FULL TEXT), expecting that older individuals—having longer work histories—would command longer descriptions. The model yields a surprising null result, showing a flat, statistically insignificant correlation across all age cohorts.
- *Sociocultural Interpretation:* Rather than concluding that age was irrelevant to enslavers, the researcher uses this biography to reframe the output. The text length of an advertisement was strictly constrained by line-cost pricing models implemented by antebellum newspaper publishers. Advertisers condensed descriptions of older individuals into concise, highly standardized negative markers (e.g., "to serve until 30 years old") to minimize advertising costs while fulfilling the legal requirement to specify

terms of service. The null result is an artifact of newspaper business practices, not an indicator of social indifference toward age.

Hypothetical Scenario B: An Explicit Distributional Anomaly

- *Plausible Computational Output:* A spatial clustering algorithm reveals an abrupt, anomalous spike in advertisements for temporary labor hires (Ad Type = "hire") concentrated in a single county during the late autumn months, contrasting sharply with steady sale volumes elsewhere.
- *Sociocultural Interpretation:* A purely technical interpretation might flag this as a data-entry duplicate or a localized economic panic. However, the context biography directs the researcher to examine regional agricultural cycles. As noted in Stephen D. Behrendt's *"Seasonality in the Trans-Atlantic Slave Trade,"* agricultural demands tightly dictated labor contracts. In regions transitioning to mixed grain farming, the late autumn marked the completion of the harvest, a period when enslavers rented out excess labor to minimize maintenance costs over the winter. The anomaly reflects a historically meaningful seasonal labor strategy rather than a data collection error.

Final Evaluative Safeguards

1. **Unresolvable Questions:** This corpus cannot reliably answer biographical questions concerning the lifetime geographic trajectories or family lineages of enslaved individuals due to systemic anonymization.
2. **Accounting for Inherited Biases:** Researchers must interpret all statistical clusters of skills or physical descriptions as commercial market claims made by subscribers, rather than objective descriptions of the enslaved population.
3. **Distinguishing Process Artifacts from Historical Patterns:** A sudden drop in advertisement volume in a specific county's newspaper index during a given year must be cross-referenced with newspaper operation histories before it is interpreted as a local decline in slave trafficking; it frequently reflects independent historical disruptions, such as press failures, paper shortages, or fires at the publisher's office.